

# Rahul Bagul

Chicago, IL, USA

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## Summary

Software engineer with 3+ years of hands-on experience designing and implementing scalable systems using **Java**, **ReactJS**, **Python** and **NodeJS**. Built full-stack applications including RESTful APIs, microservices, and data-engineering pipelines, and proficient in C++, MySQL, AWS, and modern system-design patterns, with a solid foundation in data structures and multithreading.

## Education

### Purdue University

Master of Science in Computer Science GPA: 3.9/4.0

Aug 2023 – May 2025

Indiana, USA

### Savitribai Phule Pune University

Bachelor of Engineering in Information Technology GPA: 8.55/10

Aug 2017 – May 2021

Pune, India

## Technical Skills

**Programming Languages:** Java, Python, C++, JavaScript, TypeScript, SQL, HTML/CSS

**Frameworks:** Node.js, Spring Boot, ReactJS (Context, Redux, Jest), JDBC

**Databases:** MySQL, PostgreSQL, Redis, Snowflake

**Cloud and DevOps:** AWS (EC2, S3, RDS), GCP, Salesforce, Docker, Git, Firebase, Linux

**Skills:** Design Patterns, Multithreading, Object-Oriented Design (OOD), Data structures, Micro-services, Machine Learning(ML), LLM/AI

## Work Experience

### Walmart Global Tech

June 2025 – Present

Software Engineer

Remote, USA

- Maintain and develop high-throughput **Node.js microservices** for real-time financial event processing, consuming from 3 concurrent Kafka streams (*Walmart Pay*, *Accounts*, *Walmart Plus*) and reporting data to the ONE Pay API.
- Designed and implemented a new metrics tracking service in the Omni Financial Event Service to monitor key KPIs such as request volume, error rates, processing time, resource usage, and memory consumption; built dashboards with alerting capabilities, and deployed using Kubernetes.

### Snowflake

May 2024 – Aug 2024

Data and AI Intern

San Mateo, USA

- Developed a full-cycle project to automate test class generation (73% accuracy) using **Python**, focusing on robust system design to minimize security vulnerabilities via LLM fine-tuning and RAG.
- Enhanced error detection accuracy by 16% by leveraging efficient **algorithms and data handling with ML models** (LLaMA, Mistral) for log analysis.

### Snowflake

June 2021 – July 2023

Software Engineer

Pune, India

- Engineered data integration between Salesforce and external systems, employing **APIs (REST, SOAP) with OAuth 2.0 and JWT** security, delivering a 21% efficiency increase and heightened security.
- Refactored backend logic using **Java OOP principles, SOQL** indexing, and leveraged bulk data patterns (batch jobs, queueables) to reduce query time, optimizing performance across 10K+ daily transactions and 1M+ records.
- Enhanced the UI system using modern **JavaScript and CSS** frameworks to align with current design trends, resulting in a 30% improvement in user satisfaction and adoption by 500+ GTM business users, as evidenced by positive feedback.

## Projects

### ClassroomCoder - Coding Platform | Python, ReactJS, PostgreSQL, Docker, Django, GCP

April 2025

- Designed and built a scalable B2B platform for universities featuring real-time code execution, JWT-based auth, role-based access, async grading workflows, and system design principles for high performance and security.

### Travelopedia | ReactJS, Java, Spring Boot, MySQL, GCP, MongoDB, Docker, Redis

December 2024

- Built a scalable travel planning platform using React and Spring Boot microservices, integrating Redis caching (98% latency reduction) and deploying on GCP with Docker and Kubernetes for isolated service management.

### Mastodon TutorAI | Python, MongoDB, ChromaDB, OpenAI, GCP

August 2024

- Led a team to build an AI tutoring platform using a RAG approach with LLaMA, OpenAI embeddings, ChromaDB for semantic search, and MongoDB for structured storage, while owning system architecture and development.